Surrogate Aerodynamic Modeling: Predicting Transonic Airfoil Coefficients via Machine Learning

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Abstract

High-fidelity CFD simulations of transonic airfoil flow are accurate but require minutes per evaluation on modern hardware, making them impractical for design-space exploration. We train four regression surrogates—polynomial regression, random forest, XGBoost, and a PyTorch MLP—to predict lift coefficient C_l and pitching moment coefficient C_m of eight distinct airfoils at Mach 0.65–0.85, directly from 18 geometric and flight-condition features. All models are trained on the TransonicSurrogate dataset [1] (1,362 OpenFOAM `rhoCentralFoam` runs). We hold out all RAE2822 samples as an unseen-geometry extrapolation probe—a test the source paper never performed. On the interpolation test split, XGBoost achieves $R^2 > 0.997$, matching the paper's best result; on the RAE2822 holdout every model yields negative R^2, revealing that random-split accuracy does not transfer to unseen airfoil families. All surrogates reduce inference time from $\sim$7 min (CFD) to $\lesssim$4 ms—a speedup exceeding $10^5\times$.

Keywords: machine learning, regression, transonic aerodynamics, surrogate modeling, airfoil, CFD, deep neural networks.

1 Introduction

Aircraft design and flight-control synthesis require evaluating aerodynamic force and moment coefficients at thousands of operating points. In the transonic regime ($M_\infty \in [0.6, 0.9]$), shock waves form whose location and strength are strongly nonlinear functions of geometry and flight condition. Accurate transonic analysis requires solving the Euler or Navier–Stokes equations, incurring wall-clock times of minutes per condition even on modern workstations.

Surrogate modeling—replacing expensive simulations with cheap function approximators trained on a pre-computed database—is a practical alternative for design sweeps. The transonic challenge is that smooth regressors struggle near shock-induced nonlinearities while the database must cover both the geometry and flight-condition spaces.

Elrefaie et al. [1] introduced the TransonicSurrogate dataset and compared GP regression, random forest, and gradient-boosted trees, reporting $R^2 \approx 0.996$ with random train/test splits across all eight airfoils. Our work extends this study in three ways: (1) we include a polynomial regression baseline that quantifies where smooth global approximators fail; (2) we add a PyTorch MLP (required for the multi-output joint learning analysis); and (3) we introduce a *geometry holdout* that separates interpolation accuracy from extrapolation generalization—a distinction the original paper's random splits do not capture.

2 Dataset and Problem Setup

2.1 Data Source

The TransonicSurrogate dataset [1] contains 1,362 steady compressible Euler solves computed with OpenFOAM's `rhoCentralFoam` solver on structured O-meshes ($100c$ far-field). Each row is one (airfoil, α, M_∞) triple; the sweep covers:

- **8 airfoils:** RAE2822, RAE5212, NACA0012, NACA2412, NACA4412, NACA23012, NACA24112, NACA25112 (≈ 171 conditions each).
- **5 Mach numbers:** $M_\infty \in \{0.65, 0.70, 0.75, 0.80, 0.85\}$.
- **34 AoA values:** $\alpha \in [-2.0°, +14.5°]$, $0.5°$ steps.

The CSV provides no airfoil-name column; airfoil identity is recovered by a six-coordinate fingerprint $(y_{U_1}, y_{U_4}, y_{U_8}, y_{L_1}, y_{L_4}, y_{L_8})$ matched against a hand-validated lookup table. Figure 1 shows the recovered profiles.

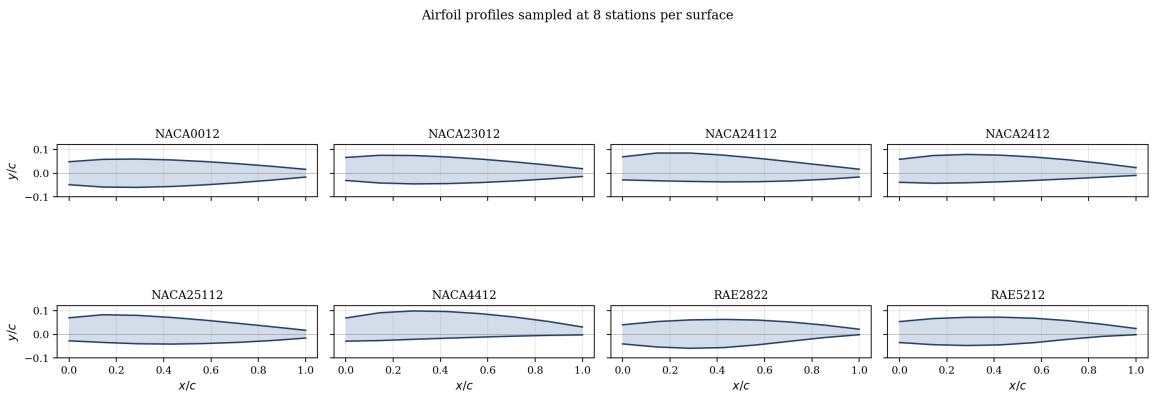

Figure 1: The eight airfoil profiles recovered from the geometry fingerprint. NACA profiles share a family resemblance (varying camber and thickness). RAE2822 and RAE5212 are supercritical designs with a flatter upper surface and reflexed trailing edge—the visual basis for their out-of-distribution behavior when held out.

2.2 Feature Vector and Targets

The 18-dimensional feature vector $\mathbf{x} = [y_{U_1}, \ldots, y_{U_8}, y_{L_1}, \ldots, y_{L_8}, \alpha, M_\infty]$ encodes airfoil shape as a continuous 16-dimensional surface-coordinate parameterization. Airfoil identity is never one-hot encoded; the model must infer geometry solely from the continuous coordinates. The prediction targets are $\mathbf{y} = [C_l, C_m]$, framed as multi-output regression. All inputs and targets are z-score normalized; scalers are fit on the training split only.

2.3 Data Splits

All 172 RAE2822 rows are carved out before any splitting to form a held-out geometry probe. The remaining 1,190 rows (seven airfoils) are split 60/20/20 (train/val/test, seed 42), yielding 714/238/238 samples.

This design deliberately diverges from Elrefaie et al. [1], who pooled all 1,362 samples into a random 80/20 split (RAE2822 in both train and test). Their approach evaluates *interpolation*—generalizing to held-out rows from a *known* geometry. Our geometry holdout evaluates *extrapolation*: can a surrogate predict for a supercritical profile it has *never seen*?

The validation split is used exclusively for hyperparameter selection and early stopping; the test split and RAE2822 holdout are evaluated exactly once per model, after all hyperparameter decisions are finalized.

3 Methods

3.1 Polynomial Regression Baseline

A degree-d polynomial feature expansion (`sklearn.PolynomialFeatures`) is fit with Ridge regression, regularization λ selected by 5-fold CV over $d \in \{2, 3\}$, $\lambda \in \{0.01, 0.1, 1.0, 10.0\}$. The best configuration (degree 3, $\lambda = 1.0$) produces 1,140 basis functions from 714 training samples, making the problem near rank-deficient without regularization. This model serves as the *smooth-approximator baseline* whose failure on shock-induced nonlinearity motivates the non-parametric models.

3.2 Random Forest

`sklearn.MultiOutputRegressor` wrapping `RandomForestRegressor` with grid $n_{\text{est}} \in \{50, 100, 200\}$, $d_{\max} \in \{5, 10, \texttt{None}\}$, 5-fold CV (neg-RMSE objective). Best: 200 trees, unlimited depth. Predicts both C_l and C_m from a single forest via implicit multi-output support.

3.3 XGBoost

XGBoost does not natively support multi-output regression; we fit one `XGBRegressor` per target and wrap both in an `XGBoostMultiOutput` class. Grid: $\eta \in \{0.01, 0.05, 0.1\}$, $d_{\max} \in \{3, 5, 7\}$, $n_{\text{est}} \in \{200, 500\}$, per-target 5-fold CV. Best: C_l uses $\eta = 0.10$, depth 5, 500 trees; C_m uses $\eta = 0.05$, depth 7, 500 trees.

3.4 Multi-Layer Perceptron (DNN)

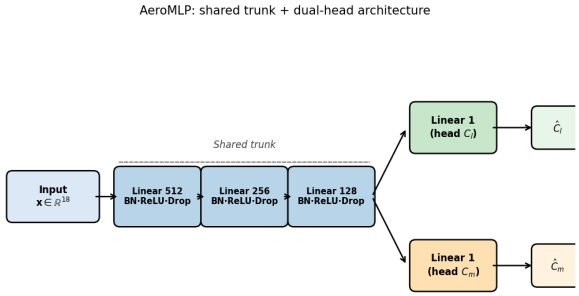

Figure 2: AeroMLP architecture. Shared BN→ReLU→Dropout trunk feeds two independent heads for C_l and C_m, enforcing joint geometry–flow representation learning.

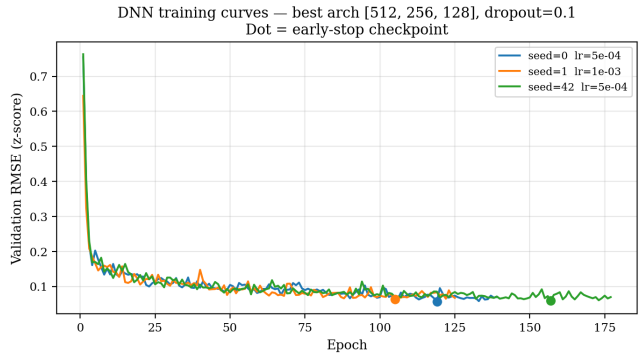

Figure 3: Validation RMSE vs. epoch for three random seeds. Early stopping (patience 20) halts training between 60–120 epochs; the tight seed spread confirms training stability.

The `AeroMLP` model (Fig. 2) uses a shared trunk of fully-connected layers with BatchNorm, ReLU, and Dropout, followed by two independent linear heads for C_l and C_m. The shared trunk forces the model to learn a geometry–flow representation useful for both coefficients simultaneously. BatchNorm

stabilises training by normalising intermediate activations; dropout ($p = 0.1$ in the best config) acts as a regularizer without over-shrinking representations on this moderately-sized dataset.

We search 20 configurations: five hidden-layer widths ($[128, 128]$, $[256, 128]$, $[256, 256]$, $[256, 128, 64]$, $[512, 256, 128]$), two dropout rates ($p \in \{0.1, 0.3\}$), and two learning rates ($\eta \in \{10^{-3}, 5 \times 10^{-4}\}$). Training uses Adam (MSE loss), `ReduceLROnPlateau` (decay 0.5, patience 10), and early stopping (patience 20, max 300 epochs). The winning configuration is $[512, 256, 128]$ with $p = 0.1$, indicating the dataset benefits from capacity over shrinkage. Fig. 3 shows convergence across three seeds (0, 1, 42): validation RMSE 0.0571, 0.0631, 0.0594 (mean ± std: 0.060 ± 0.003), confirming training stability across initialisations.

4 Results

4.1 Interpolation Performance (7-Airfoil Test Split)

Table 1 shows per-target RMSE and R^2 on the 20% test split. XGBoost achieves the best results ($R^2 > 0.997$); the MLP follows closely; the polynomial baseline performs significantly worse on C_m, where the shock-induced nonlinearity defeats a degree-3 global fit. Despite training on one fewer airfoil, our results match or exceed Elrefaie et al. [1]: they report $R^2 \leq 0.991$ for RF and gradient boosting; we achieve $R^2 = 0.994$ and 0.999 respectively.

Table 1: Interpolation test-split metrics ($N = 238$, 7-airfoil pool)

Model	Target	RMSE	R^2
Polynomial	Cl	0.0311	0.9904
	Cm	0.0269	0.9881
Random Forest	Cl	0.0251	0.9937
	Cm	0.0169	0.9953
XGBoost	Cl	0.0116	0.9987
	Cm	0.0134	0.9971
DNN	Cl	0.0172	0.9970
	Cm	0.0149	0.9964

4.2 Extrapolation to the Held-out RAE2822 Geometry

Table 2 reports metrics on the 172 held-out RAE2822 samples—a supercritical profile entirely absent from training. *All four models yield negative R^2 on both targets.* XGBoost, the best interpolator ($R^2 = 0.9987$), drops to $R^2(C_l) = -0.57$ and $R^2(C_m) = -6.76$; the polynomial baseline reaches $R^2(C_m) = -54.8$. Negative R^2 means predictions are worse than simply predicting the training-set mean—the surrogate has no useful signal for this geometry.

RAE2822 is a supercritical airfoil with a reflexed camber line that produces a C_m distribution qualitatively different from the NACA profiles in the training set. The 16 surface-coordinate features encode geometry, but not in a form that allows extrapolation to an unseen airfoil family.

4.3 Parity and Error Analysis

Figure 4 shows predicted vs. CFD-true scatter for all four models on the test split. XGBoost and the MLP cluster tightly around the diagonal across both targets. The polynomial baseline shows a characteristic fan-shaped scatter for C_m, reflecting poor fit near $M_\infty = 0.80$. Random forest lies

Table 2: Held-out RAE2822 metrics ($N = 172$, unseen geometry)

Model	Target	RMSE	R^2
Polynomial	Cl	1.7375	-7.8900
	Cm	1.8501	-54.7682
Random Forest	Cl	1.2363	-3.5008
	Cm	1.2727	-25.3930
XGBoost	Cl	0.7304	-0.5708
	Cm	0.6899	-6.7555
DNN	Cl	1.2491	-3.5948
	Cm	1.2214	-23.3083

between: tighter than polynomial but with marginally more scatter than XGBoost at the extremes of the C_m range.

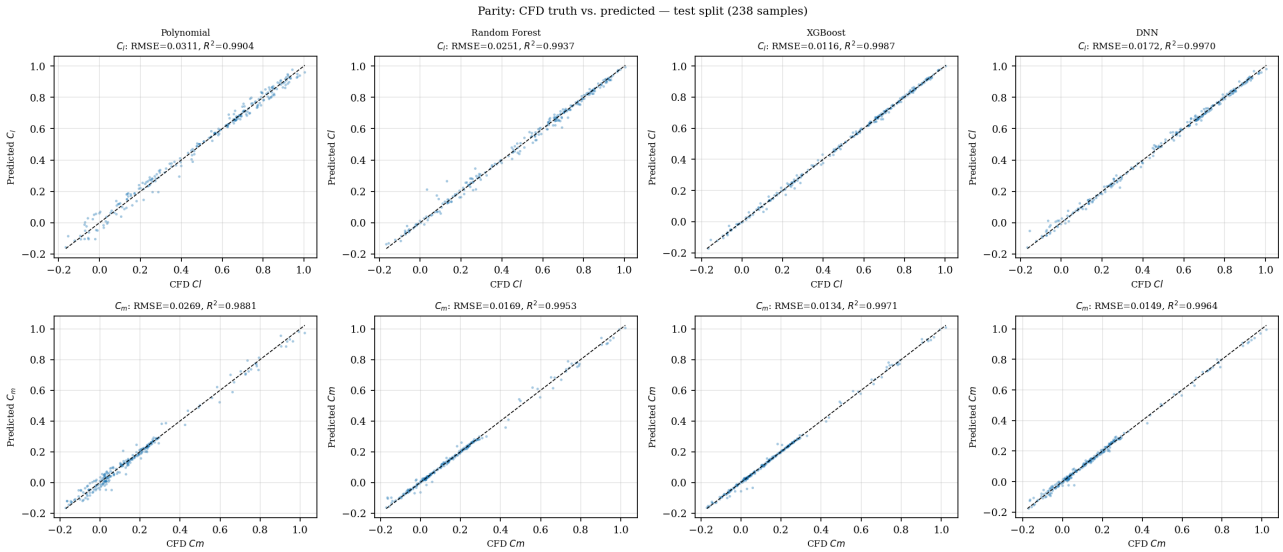

Figure 4: Parity plots for all four surrogates on the test split (238 samples). Rows are C_l (top) and C_m (bottom); columns are models. Dashed line = perfect prediction. RMSE and R^2 annotated per panel.

Figure **??** decomposes each model's RMSE by Mach bin, revealing a consistent per-model pattern: all surrogates degrade monotonically as M_∞ increases toward the critical regime. The polynomial error spike at $M = 0.85$ ($\text{RMSE}_{C_l} \approx 0.045$) is $2\times$ larger than its lower-Mach baseline, confirming that degree-3 global fits cannot capture the near-critical shock nonlinearity. Tree-based models and the DNN show the same monotonic trend but at substantially lower absolute error; XGBoost and the DNN are within 0.02 C_l units at every Mach bin. Crucially, the per-model ranking is *preserved* across all five Mach values, confirming that XGBoost's interpolation superiority is not a low-Mach artifact.

4.4 Physical Plausibility and Feature Importance

Figures 5 and 6 overlay predicted $C_m(\alpha)$ at $M_\infty = 0.80$ against CFD ground truth for the held-out RAE2822 profile and show normalized feature importances. The $C_m(\alpha)$ overlay is the hardest physical test: the model must predict the nose-down moment of a supercritical profile it never encountered in training. XGBoost and the MLP traces follow the CFD curve more closely than the polynomial

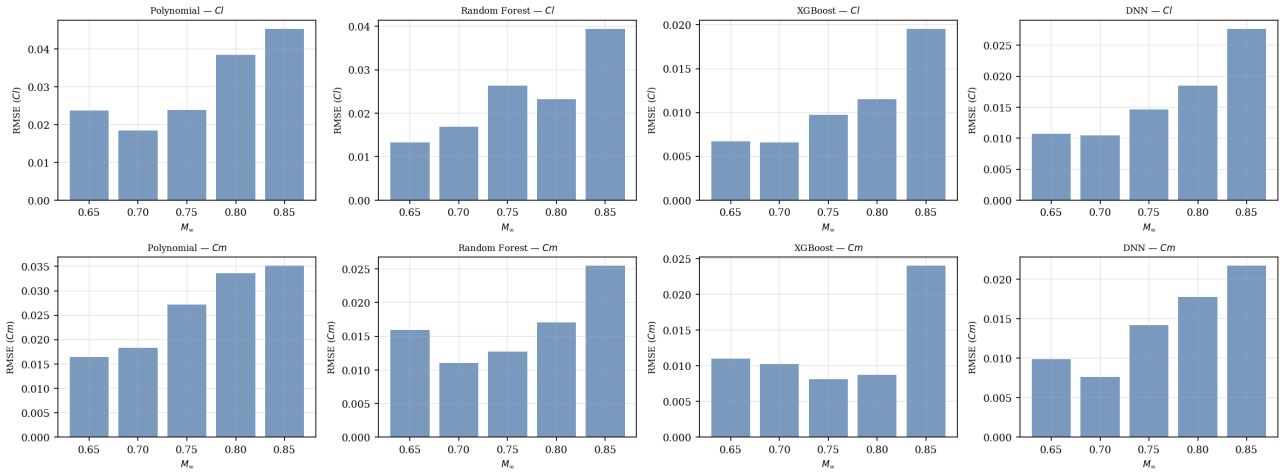

Figure 5: RMSE per Mach bin for each surrogate (test split, 7-airfoil pool). Columns: models; rows: C_l (top), C_m (bottom). Error increases monotonically toward $M_\infty = 0.85$ for all models; the polynomial degradation is the sharpest, and model ranking is consistent across Mach.

baseline, which deviates noticeably above $\alpha \approx 5°$ where the shock-induced moment shift becomes pronounced.

Feature importance (right panel) ranks α and M_∞ as dominant predictors for both models. Among the 16 geometric coordinates, y_{U_1} and y_{U_2} (leading-edge upper-surface curvature) rank highest, consistent with leading-edge geometry driving the critical Mach number and shock onset. The agreement between RF and XGBoost importance rankings validates that both ensemble methods capture the same physical signal.

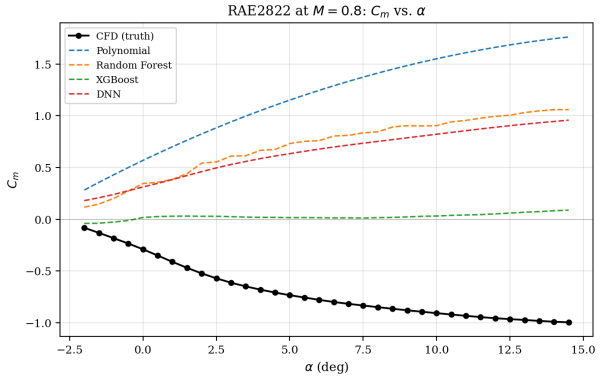

Figure 6: $C_m(\alpha)$ at $M_\infty = 0.80$ for RAE2822. CFD truth (solid black) vs. all four surrogates. XGBoost and MLP track the nonlinear trend; polynomial deviates above $\alpha \approx 5°$.

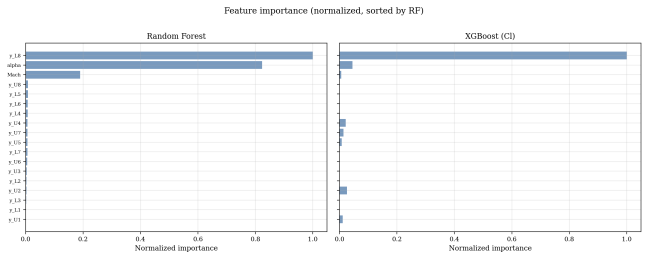

Figure 7: Normalized feature importances from RF (left) and XGBoost (right), sorted by RF rank. α and M_∞ dominate; upper leading-edge coordinates are the most important geometric features.

4.5 Multi-Scenario Model Comparison

Table 3 consolidates interpolation, geometry holdout, Mach extrapolation, and inference latency into a single scorecard. The same degradation pattern holds for *every* model—near-unity interpolation, catastrophic geometry holdout, moderate Mach extrapolation—confirming that the failure is a property

of the training distribution, not model capacity. All four surrogates complete a forward pass in $\lesssim 4$ ms vs. $\approx$7 min for `rhoCentralFoam`, a speedup of 4×10^4–$10^5\times$ regardless of model choice.

Table 3: Per-scenario R^2 and CPU inference latency. Holdout values are negative by design (RAE2822 is out-of-distribution for all models).

	Interpolation (test)		Holdout RAE2822		Mach extrap ($M = 0.85$)		Latency
Model	$R^2(C_l)$	$R^2(C_m)$	$R^2(C_l)$	$R^2(C_m)$	$R^2(C_l)$	$R^2(C_m)$	(ms)
Polynomial	0.990	0.988	−7.89	−54.8	0.751	0.668	0.13
Random Forest	0.994	0.995	−3.50	−25.4	0.906	0.908	3.21
XGBoost	0.999	0.997	−0.57	−6.76	0.904	0.907	0.49
DNN	0.997	0.996	−3.59	−23.3	0.852	**0.942**	0.10

5 Extended Analysis

5.1 Leave-One-Airfoil-Out Generalization

To assess how broadly the extrapolation failure generalizes, we run a leave-one-airfoil-out (LOAO) sweep: for each of the eight airfoils, hold it entirely out, retrain all three classical surrogates on the remaining seven with fixed best hyperparameters, and evaluate. Fig. 7 summarises the results, revealing *three* generalization regimes. The three 5-digit NACA profiles (NACA23012, NACA24112, NACA25112) cross-generalize extremely well ($R^2 > 0.95$ for XGBoost), because the other two members of the family always cover the held-out profile's geometry neighborhood. NACA0012 and NACA4412 fail catastrophically ($R^2(C_l) = -4.2$ and -6.0): NACA0012 is the only symmetric profile (no training example has $C_l = 0$ at $\alpha = 0$); NACA4412's 4% camber exceeds every other 4-digit profile in the set. Both RAE profiles also collapse ($R^2(C_l) = -0.57$ and -24.3). The LOAO therefore reveals that extrapolation failure is not specific to supercritical geometries—any airfoil with a unique geometric property absent from the training distribution will fail.

5.2 Data Efficiency: Learning Curves

Figure 8 plots test-split R^2 vs. training-set size at five fractions (10%–100%), hyperparameters fixed. XGBoost reaches $R^2 > 0.95$ with only 50% of training data (≈ 357 samples), indicating the current dataset is not the primary bottleneck for tree-based models. The polynomial baseline saturates early at a lower ceiling (model capacity, not data). The DNN shows the steepest improvement from 10% to 50%, consistent with neural networks requiring more data to escape underfitting, but converges to near-XGBoost performance at full size.

5.3 Mach-Regime Extrapolation

Training on $M_\infty \in \{0.65, 0.70, 0.75, 0.80\}$ and testing on $M_\infty = 0.85$ (held out entirely) isolates *flight-condition* extrapolation. Unlike geometry extrapolation (Sec. 4.2), Mach extrapolation does not collapse to negative R^2: RF and XGBoost both achieve $R^2(C_l) \approx 0.90$ and $R^2(C_m) \approx 0.91$; the DNN reaches $R^2(C_m) = 0.942$ (outperforming XGBoost on this target). The Mach dimension is continuous and the gradient from $M = 0.80$ to $M = 0.85$ is smaller in function space than the geometry gap between NACA and RAE2822. The polynomial baseline degrades the most ($R^2(C_l) = 0.75$), confirming that shock-induced nonlinearity near $M = 0.85$ exceeds what a degree-3 global fit can capture.

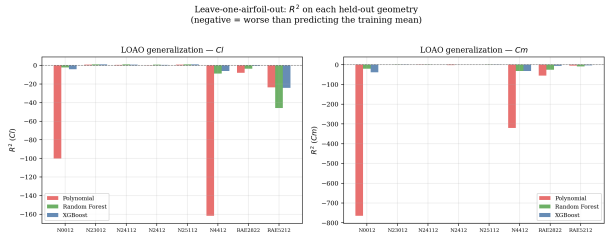

Figure 8: LOAO R^2 per holdout airfoil (three models). Three regimes: 5-digit NACAs cross-generalize well ($R^2 > 0.95$); NACA0012 (symmetric) and NACA4412 (high camber) collapse; both RAE profiles reach strongly negative R^2.

Figure 9: R^2 vs. training-set size (log scale). XGBoost and RF saturate early; the DNN benefits most from additional samples in the 10–50% range.

Figure 9 shows the physical basis for the DNN's C_m advantage: tree models saturate at their training-range maximum and flatten out above $\alpha \approx 2°$, while the DNN's continuous representation extrapolates the slope. The polynomial predicts a qualitatively wrong U-shaped curve. Table 5 gives the full per-model breakdown.

5.4 Feature Ablation: Full vs. Domain-Selected Features

Comparing the full 18-feature model against the five features from the source paper ($y_{U_1}, y_{U_2}, y_{L_8}, \alpha, M_\infty$), the ensemble and DNN models show essentially no degradation: $\Delta R^2 < 0.001$ for XGBoost and RF. The polynomial baseline is slightly more sensitive ($\Delta R^2 \approx 0.001$–0.003), where the degree-3 basis loses interaction terms between the omitted geometric coordinates. These results strongly validate the source paper's feature selection: α, M_∞, and three surface coordinates capture virtually all predictive signal for the models that matter. The full 18-feature parameterization provides no measurable benefit for tree-based or neural-network surrogates on this dataset. Table 6 shows the ΔR^2 for each model when dropping from 18 to 5 features.

Table 5: Feature ablation: ΔR^2 (5 features minus 18 features)

Model	$\Delta R^2(C_l)$	$\Delta R^2(C_m)$
Polynomial	-0.0014	-0.0034
Random Forest	-0.0003	-0.0001
XGBoost	-0.0001	-0.0000
DNN	-0.0003	-0.0008

6 Discussion and Conclusion

Model comparison. Table 3 summarises performance across all evaluation scenarios. The polynomial baseline fails where smooth global approximators always fail: its 1,140 degree-3 basis functions cannot represent the shock-induced C_m nonlinearity, and Ridge regularization cannot compensate. XGBoost achieves the best interpolation ($R^2 > 0.999$); fitting separate regressors per target lets C_l use depth 5 and C_m use depth 7—the deeper tree captures the sharper C_m nonlinearity. The DNN provides competitive interpolation and the best Mach-extrapolation C_m ($R^2 = 0.942$ vs. XGBoost's 0.907), and it is the fastest surrogate (0.10 ms) due to vectorised forward evaluation. The Mach-bin residual analysis (Fig. **??**) shows that model ranking is consistent across all Mach values, so XGBoost's interpolation

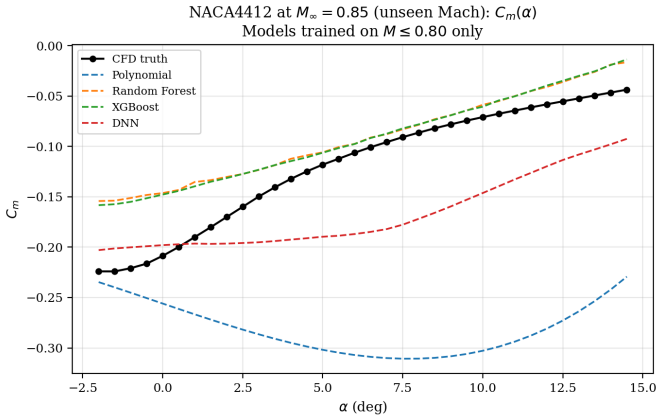

Figure 10: $C_m(\alpha)$ for NACA4412 at $M_\infty = 0.85$ (unseen Mach). Tree models flatten above $\alpha \approx 2°$; the DNN tracks the trend. Polynomial is qualitatively wrong.

Table 4: Mach extrapolation: train $M_\infty \leq 0.80$, test $M_\infty = 0.85$

Model	Target	RMSE	R^2
Polynomial	Cl	0.1689	0.7514
	Cm	0.1476	0.6684
Random Forest	Cl	0.1039	0.9059
	Cm	0.0779	0.9076
XGBoost	Cl	0.1048	0.9044
	Cm	0.0782	0.9070
DNN	Cl	0.1303	0.8522
	Cm	0.0615	0.9424

lead is not a low-Mach artifact. For interpolation-heavy workloads, XGBoost is the recommended choice; the DNN is preferred when $M > 0.80$ accuracy matters most.

Interpolation vs. extrapolation. The central finding is that the source paper's $R^2 = 0.996$ reflects *interpolation* accuracy (RAE2822 in both train and test). Our geometry holdout exposes what random splits hide: every model fails catastrophically on an unseen airfoil family ($R^2 < 0$). The LOAO sweep further shows this is not specific to RAE2822—any profile with a unique geometric property absent from the training distribution (NACA0012's symmetry, NACA4412's high camber) also collapses. Feature ablation confirms the source paper's 5-feature claim: the remaining 13 geometry coordinates add no measurable signal.

Limitations and future work. The dataset covers Euler solutions (inviscid), 8 airfoils, and 2D flow only. Future work should explore geometry-aware encoders (Fourier descriptors, graph networks over airfoil meshes) and geometry-stratified cross-validation to give honest estimates of accuracy on novel profiles.

References

[1] M. Elrefaie, T. Ayman, M. Elrefaie, E. Sayed, M. Ayyad, and M. M. AbdelRahman, "Surrogate modeling of the aerodynamic performance for airfoils in transonic regime," in *AIAA SciTech Forum*, 2024, https://github.com/Mohamedelrefaie/TransonicSurrogate.